

MPG No. 14.5.52 $\bar{4}2'm'$ (-42'm' setting) [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = M_{pz}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ 0 & -T_{xxz} & 0 \\ T_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = T_{fbz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ 0 & \frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = 2M_{dxy}^{(1)}$$

$$\frac{T_{xxz}}{2} = M_{dxy}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ T_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = M_{dxy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxxy} & -T_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = T_{gaz}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & 0 & 0 \\ -T_{xxxy} & T_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = -2M_{faz}^{(1)} + 2M_{pz}^{(1)}$$

$$T_{yzzx} = -4M_{faz}^{(1)} - M_{pz}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -M_{pz}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxx} \\ 0 & 0 & T_{xxxx} \\ 0 & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = -M_{faz}^{(2)} + M_{pz}^{(3)}$$

$$T_{zzxy} = 2M_{faz}^{(2)} + M_{pz}^{(3)} + 2M_{pz}^{(4)}$$

$$T_{yzzx} = -M_{faz}^{(2)} + M_{pz}^{(4)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyyz} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -M_{faz}^{(3)} + M_{pz}^{(6)}$$

$$T_{xyxx} = -M_{faz}^{(3)} + M_{pz}^{(5)}$$

$$T_{xyyz} = 2M_{faz}^{(3)} + M_{pz}^{(5)} + 2M_{pz}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxx} & 0 & 0 & T_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & -T_{xxyx} & 0 & 0 & -T_{xxxx} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & -T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzxz} & 0 & 0 & -T_{yzzx} & 0 & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = -2M_{faz}^{(1)} - M_{faz}^{(2)} + 2M_{pz}^{(1)} + M_{pz}^{(3)} + T_{gaz}^{(1)}$$

$$T_{xxyx} = -2M_{faz}^{(1)} + M_{faz}^{(2)} + 2M_{pz}^{(1)} - M_{pz}^{(3)} + T_{gaz}^{(1)}$$

$$T_{zzxy} = 2M_{faz}^{(2)} + M_{pz}^{(3)} + 2M_{pz}^{(4)}$$

$$T_{yzzx} = -4M_{faz}^{(1)} - M_{faz}^{(2)} - M_{pz}^{(1)} + M_{pz}^{(4)}$$

$$T_{yzxz} = -4M_{faz}^{(1)} + M_{faz}^{(2)} - M_{pz}^{(1)} - M_{pz}^{(4)}$$

$$T_{xyxx} = 2M_{faz}^{(1)} - 2M_{pz}^{(1)} + T_{gaz}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzxz} & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyyz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -M_{faz}^{(3)} - M_{pz}^{(2)} + M_{pz}^{(6)}$$

$$T_{yzxz} = -M_{faz}^{(3)} + M_{pz}^{(2)} + M_{pz}^{(6)}$$

$$T_{xyxx} = -M_{faz}^{(3)} + M_{pz}^{(5)}$$

$$T_{xyzz} = 2M_{faz}^{(3)} + M_{pz}^{(5)} + 2M_{pz}^{(6)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & 0 \\ -T_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxxy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = T_{gaz}^{(1)}$$